

Beyond Keywords: A Hybrid Neural Semantic Approach for Coffee Recommendation

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Abstract

This project implements a hybrid recommendation engine to address the disconnect between informal user preferences and professional coffee cupping notes. Our approach combines regular expressions to enforce objective constraints—such as price, origin, and roast level—with a Sentence-BERT (SBERT) layer for semantic flavor mapping. We evaluated the system across three realistic scenarios: integrated constraints, lexical noise (typos), and complex sensory descriptions. Experimental results show that SBERT outperforms a TF-IDF baseline, particularly in descriptive queries where it achieved a 187.48% improvement in similarity scores by correctly mapping metaphorical intent (e.g., “like a garden”) to specific floral descriptors. The results show that the sharp increase in retrieval confidence did not result in a corresponding jump in BERTScore. This indicates that capturing the underlying sensory profile is more essential for coffee recommendation than the literal token overlap tracked by BERTScore.

1 Introduction

In the specialty coffee industry, flavor descriptions are often highly subjective and complex. While professionals use standardized cupping notes, ordinary consumers tend to use natural language, which is often vague or contains spelling errors. Traditional recommendation systems rely on keyword-based matching, such as TF-IDF. However, these models suffer from the “vocabulary gap”—they fail to capture the underlying meaning of words and are highly sensitive to typos.

This project aims to bridge this gap by developing a Hybrid Recommendation Architecture. We first established a Baseline Model using TF-IDF, then transitioned to an Advanced Model based on Sentence-BERT (SBERT). This approach allows for semantic retrieval, enabling the system to understand that terms like “almond” and “nutty” are con-

ceptually related. Additionally, we integrated Regular Expressions (Regex) to handle hard constraints like price and roast level. To evaluate the system’s performance, we adopted a dual approach combining human qualitative assessment and BERTScore.

2 Theory

The project transitions from keyword-based retrieval to semantic vector spaces. Unlike traditional TF-IDF, which relies on term frequency and ignores context, our approach uses **Sentence-BERT (SBERT)**. As introduced by (Reimers and Gurevych, 2019a), SBERT employs a Siamese Network architecture to generate fixed-size dense embeddings for sentences.

By mapping descriptions into a continuous vector space, the system calculates **cosine similarity** to identify coffees that are semantically related (e.g., “jasmine” and “floral”) even without literal word overlaps. The cosine similarity between a query vector **A** and a description vector **B** is defined as:

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (1)$$

where n represents the dimensionality of the embedding space.

For evaluation, in addition to **human qualitative assessment**, we apply **BERTScore**, which uses contextual embeddings to map tokens between the user query and the recommended description.

3 Data

The dataset used in this project is sourced from a collection of coffee reviews on Kaggle (Schmoyote, 2022), which was originally scraped from *CoffeeReview.com*. The raw dataset contains 2,095 samples, representing various specialty coffees with features such as *name*, *price*, *roaster*, *roast level*, *origin*, *rating* and *description*.

The *description* field is the core of our semantic analysis, consisting of **professional cupping notes** and **sensory descriptors** provided by expert reviewers. These notes provide a standardized organoleptic evaluation of the coffee's aroma, acidity, body, and flavor nuances.

For preprocessing, we conducted several data cleaning steps to ensure model robustness. This included the removal of entries with missing values (NaNs) and the filtration of invalid data points. Additionally, geographic origins were standardized, and pricing information was normalized to a consistent *USD per 100g* scale to facilitate accurate hard-filtering during the recommendation process.

4 Method

The primary objective of the proposed system is to recommend coffee beans that best match user-specified preferences, including origin, roast level, roaster, price, and desired flavor profiles, by using historical professional cupping data.

In our recommendation system, constraints such as origin, roast level, roaster, and price are treated as objective criteria that can be implemented through **hard filtering**. We utilize **Regular Expressions (Regex)** to parse structured information from the user's natural language input, allowing the system to execute precise filtering based on formal language theory and string matching principles (Aho and Ullman, 1972).

Regarding the flavor profile, which is inherently subjective and linguistically complex, discrepancies often exist between expert cupping notes and amateur user descriptions. To bridge this gap, we introduce a dual-methodology approach: a keyword-based retrieval system using **TF-IDF** (Baseline) and a neural semantic search system using **Sentence-BERT (SBERT)** (Advanced). Both methodologies are integrated with the aforementioned hard filters to ensure that the recommendations satisfy both the categorical constraints and the nuanced sensory preferences of the user.

4.1 Baseline Model: Hybrid TF-IDF Search

The baseline recommendation system combines deterministic hard filtering for categorical constraints with soft matching for flavor profiles using vector similarity. The implementation consists of the following four main stages:

1. Pre-processing and Feature Extraction Before retrieval, the textual descriptions (`desc_1`)

are pre-processed through a function called `simple_preprocess`. This function converts text to lowercase and removes non-alphanumeric characters using regular expressions to ensure a clean token stream. For vectorization, we utilize the `TfidfVectorizer` from the *scikit-learn* library. We apply standard English stop-word removal and set `min_df=1` to ensure all unique flavor descriptors in the dataset are captured in the sparse matrix (`tfidf_matrix`).

2. Natural Language Intent Parsing To handle structured requirements like price and roast level from a natural language query, the system uses custom extraction functions:

- **Price Extraction:** The `extract_price_from_query` function applies regex patterns to identify keywords such as "under" or "budget of" followed by numerical values, as well as numeric values followed by currency units.
- **Roast Level Extraction:** The `extract_roast_from_query` function maps the query against a dictionary of predefined roast levels (e.g., Light, Dark) using regular expressions.

3. Constraint-Based Hard Filtering Within the `baseline_search` function, the parsed intent is used to generate a Boolean mask against the cleaned dataset. The system iterates through predefined lists of roasters and origins. If a matching entity is identified within the query, it is added to the active filter set. The system uses this mask to filter the dataset. For example, if the query includes "under 20 dollars," only records where the `100g_USD` column satisfies the condition are passed to the semantic scoring stage.

4. Semantic Soft Matching and Ranking Once the hard constraints step are implemented, the system performs textual similarity matching:

- **Cosine Similarity:** The cleaned user query (with price and roast terms removed to reduce noise) is transformed into the TF-IDF space. We calculate the cosine similarity between the query vector and the subset's TF-IDF vectors.
- **Hierarchical Sorting:** The final recommendations are ranked using a two-tier strategy. Results are sorted primarily by their similarity score, followed by their professional rating

as a secondary criterion. In cases where no direct flavor match is identified (i.e., a similarity score of zero), the system defaults to ranking based on ratings alone. This fallback mechanism ensures that the user receives high-quality suggestions that still meet all hard constraints, even when specific semantic matches are unavailable.

4.2 Advanced Model: Hybrid SBERT Semantic Search

To overcome the inherent limitations of keyword-based retrieval, we applied an advanced recommendation engine using **Sentence-BERT (SBERT)**.

1. Neural Embedding and Pre-computation

The system uses the `all-MiniLM-L6-v2` model, a light-weight Sentence Transformer pre-trained on large-scale datasets (Reimers and Gurevych, 2019b). This model is based on the **MiniLM** architecture, which applies deep self-attention distillation to achieve high performance with significantly reduced computational parameters (Wang et al., 2020). To ensure real-time query performance, we perform an offline encoding of the entire dataset. Each professional cupping description is mapped into a 384-dimensional dense vector (embedding) that captures its latent sensory features. These embeddings are saved as a tensor file (`coffee_embeddings.pt`) to be loaded during inference.

2. Integrated Retrieval Logic The `sbert_search` function keeps the hybrid architecture. It shares the same Regex-based extraction logic (Aho and Ullman, 1972) to isolate categorical constraints. Once hard filters are applied, the system isolates a subset of the pre-computed embeddings for the final semantic similarity calculation.

3. Semantic Mapping and Ranking The recommendation engine processes sensory intent using the following mechanisms:

- **Query Encoding:** The cleaned query is encoded into a dense vector using the same SBERT model.
- **Vector Similarity:** We compute the **Cosine Similarity** between the query vector and candidate embeddings using `util.cos_sim`.

- **Lexical Robustness and Contextual Association:** SBERT shows better robustness performance to lexical noise when provided with contextual cues. Our experiments show that a multi-word query such as “floral jasmin” successfully maps to the professional description “jasmine” by using the semantic relationship between tokens. Furthermore, the model can associate abstract concepts like “summer” or “refreshing” with professional sensory notes such as “bright acidity” or “citrus-toned” (Reimers and Gurevych, 2019a).

5 Results and Discussion

We evaluate the recommendation engine performance by comparing the Baseline (TF-IDF) and Advanced (SBERT) models across three distinct scenarios. The evaluation uses two metrics: the internal **Similarity Score** (measuring retrieval confidence) and **BERTScore F1** (Quantifying semantic alignment between model output and user intent).

5.1 Scenario 1: Integrated Query

This scenario tests the system’s ability to process complete requirements consisting of roast level, roaster, origin, price, and sensory notes.

Query: “*Light roast coffee from Ethiopia under 20 dollars with bright citrus and floral notes.*”

Metric (Top-5)	Baseline	Advanced	Improv.
Mean Similarity	0.1363	0.4382	+221.6%
Mean BERTScore	0.8385	0.8352	-0.39%

Table 1: Quantitative results for Scenario 1 (Integrated Query).

Case Study: Both models successfully executed the hard filters for Ethiopia, Light roast, and price constraints. However, their semantic interpretation differed:

- **Baseline (TF-IDF):** The top result was *Ethiopia Yirgacheffe G1* (Sim: 0.2901). Its description includes “citrus-toned finish,” but the score is penalized by the sparse nature of keyword matching.
- **Advanced (SBERT):** The top result was *Karibu Espresso Blend* (Sim: 0.4555). SBERT more effectively mapped the user’s abstract intent to specific sensory descriptors, such as linking the query term “bright” to the professional cupping note “crisply sweet-tart”.

The slight decrease in BERTScore (-0.39%) might indicate that while SBERT captures the latent sensory profile more accurately, it places less weight on the literal token overlap that BERTScore emphasizes.

5.2 Scenario 2: Lexical Robustness

This scenario evaluates the system’s tolerance for non-standard user input, such as typos and informal phrasing, which are common in real-world search queries.

Query: “*floral jasmin*” (misspelled *jasmine*).

Metric (Top-5)	Baseline	Advanced	Improv.
Mean Similarity	0.3833	0.5077	+32.45%
Mean BERTScore	0.7810	0.7876	+0.85%

Table 2: Quantitative results for Scenario 2 (Lexical Robustness).

Case Study Analysis: The experimental results show that SBERT possesses better lexical robustness performance compared to the TF-IDF baseline.

- **Baseline (TF-IDF):** The model only successfully matched the token “floral,” leading to recommendations like *Finca Lerida Gesha Natural*. While this bean is “floral-toned,” its primary notes are “Magnolia” and “almond,” failing to capture the specific jasmine intent due to the spelling mismatch.
- **Advanced (SBERT):** Despite the misspelling of “jasmin,” the SBERT model used **Contextual Association** to correctly match the search intent with records containing “jasmine”. For example, the top result *Panama Finca Mi Finquita Geisha* explicitly identifies “Jasmine” in its professional cupping notes.

By using the semantic relationship between the anchor word “floral” and the noisy token “jasmin,” SBERT maintained high retrieval confidence (0.5077) and improved the objective semantic alignment (BERTScore) by 0.85% over the baseline.

5.3 Scenario 3: Descriptive Query (Sensory Profile Synthesis)

This scenario looks at how well the models understand the overall flavor described in long, everyday sentences instead of just searching for individual words.

Query: “*very floral like a garden with some citrus fruit mixed in.*”

Metric (Top-5)	Baseline	Advanced	Improv.
Mean Similarity	0.2243	0.6447	+187.48%
Mean BERTScore	0.8190	0.8207	+0.21%

Table 3: Quantitative results for Scenario 3 (Descriptive Query).

Case Study Analysis: SBERT shows its most significant performance gain in this scenario, with a 187.48% increase in retrieval confidence.

- **Baseline (TF-IDF):** The model is highly susceptible to **Keyword Dilution**. It prioritized literal matches for non-flavor tokens such as “garden” and “mixed,” resulting in recommendations like *Ecuador La Hormiga “Naty”*, which contains “sweet garden herbs”. This demonstrates a failure to distinguish between metaphorical language and core flavor attributes.
- **Advanced (SBERT):** The model effectively understood the sensory profile, ignoring the noise of the “garden” metaphor to focus on the core “floral” and “citrus” description. Its top results, such as *Ethiopia Dumerso Light Roast*, feature high-quality descriptors like “complex flowers (lily to lavender)” and “citrusy and floral”.

The results indicate that the SBERT-based model effectively maps holistic, natural language descriptions to expert terminology. Compared to the keyword-based baseline, this approach achieves a more consistent semantic alignment, yielding a Mean Similarity of 0.6447.

6 Conclusion

6.1 Summary of Analysis

This project evaluated the effectiveness of a coffee recommendation engine by comparing a traditional lexical-based model (**TF-IDF**) with a semantic-based model (**Sentence-BERT**). The evaluation across three distinct scenarios highlights the following observations:

- **Effectiveness in Intent Matching:** The results show that the SBERT-based model consistently produces higher similarity scores across all test cases. Particularly for long-form sensory descriptions, the similarity score

increased by 187.48%, suggesting that the model is more effective at identifying the intended flavor profiles than the baseline.

- **Analysis of Evaluation Metrics:** We noticed a clear gap between the significant increase in internal similarity scores and the relatively small gains in BERTScore (ranging from 0.21% to 0.85%). This indicates that the neural model prioritizes the latent sensory characteristics of the coffee rather than relying on literal word-for-word alignment.
- **Resilience to Input Noise:** The experiment also shows that the SBERT model is more resilient to input noise, such as typos. For instance, it successfully retrieved relevant results for the misspelled query “jasmin,” whereas the baseline model failed because it requires exact string matches to function effectively.

6.2 System Evaluation and Implementation

The project successfully addressed the challenge of bridging the gap between amateur natural language and professional cupping notes through a **hybrid architecture**:

- **Hybrid Filtering:** By using regular expressions to extract hard constraints such as price, roast level, and origin, the system ensured that objective requirements were strictly met.
- **Semantic Mapping:** SBERT was used to handle soft sensory preferences, accurately mapping user terms like “bright citrus” to professional descriptors like “crisply sweet-tart”.
- **Contextual Synthesis:** The system effectively ignored non-flavor noise in long-form queries, preventing the model from being misled by metaphors like “garden” or “mixed”.

6.3 Future Directions

To further enhance precision, the SBERT model could be fine-tuned on a specialized coffee terminology dataset. This would allow the model to learn the unique semantic nuances of sensory vocabulary, further reducing the semantic drift observed in specialized cupping notes. Furthermore, the system’s utility could be expanded by implementing automatic currency identification and conversion, which would enable the price filter to process diverse global currencies beyond the current USD standard.

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